# Assignment 2 - Probability

## Due: October 15 at 10pm

*Submit your assignment by the due date to Gradescope. Recall that the rules of academic integrity for this class allow you to discuss the questions with your peers, but the write up must be your own. You* ***may not use any other sources*** *such as the internet or other textbooks to find solutions to these questions. Your answers will be graded both on their final solution as well as the work you show to achieve that solution. Note that a subset of the questions will be graded. By submitting your assignment you agree to abide by these rules.*

1. The weather forecast suggests that the probability of rain is 90% today and 55% tomorrow. Assuming that rain today and rain tomorrow are independent events, what is the likelihood that it rains at least one day?
2. Two cards are drawn from a deck of cards with replacement (you may assume that the cards in a suit are {2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A}. Find the probability that:
   (a) The first card is a King and the second card is black.
   (b) Neither card has a value from $\{J, Q, K, A\}$.
3. In the card game Black Jack your opponent receives a face-down card and a face-up card. In particular, you can see your opponents face-up card. This lends itself to *conditional probability*. For each question specify events $E_1$ and $E_2$ so that the give question is asking you to determine $P(E_1|E_2)$ and then answer the question.
   (a) What is the probability of that the face-down card is a 10, jack, queen or king, given that you see his face-up card is an ace?
   (b) What is the probability of that the face-down card is a 10, jack, queen or king, given that you see his face-up card is an ace and you know your own hand consists of a 9 and a 10?
4. Suppose that you buy a lottery ticket containing $k$ distinct numbers from among $\{1, 2, \ldots, n\}$ where $1 \leq k \leq n$. To determine the winning ticket, $k$ balls are randomly drawn without replacement from a bin containing $n$ balls numbered $1, 2, \ldots, n$.
   (a) What is the probability that at least one of the numbers on your lottery ticket is among those drawn from the bin?
   (b) Prizes start when you have at least $\frac{k}{2}$ (ie. half) numbers in common with the drawn balls. Express the probability that you win something. You may find it helpful to use summation notation.
5. The Toronto Blue Jays are going to the playoffs! Let's consider a best of three playoff series. If the Jays win every game they play with probability $\frac{3}{5}$, then what is the probability that the Jays will win their best of three series? A best of three series means the first team to win 2 games wins the series.
6. A fair coin is tossed until either a head comes up or four tails are obtained. What is the expected number of tosses?
7. Suppose that a certain tech company sources their chips from three different factories, $F_1, F_2$ and $F_3$. Suppose the percentages of each shipment that are defective are $10\%, 8\%$ and $3\%$ respectively. Factory $F_1$ supplies 20%, $F_2$ supplies 30% and $F_3$ supplies 50% of the chips to the tech company. Suppose quality control selects a chip at random and it is defective. What is the probability the chip came from factory $F_2$?
8. A treasure chest contains 25 gold coins and 15 silver coins. Two are chosen at random one after the other, without replacement (ie, once you take the first one out, there are now 39 coins in the chest).
   (a) Use a tree diagram to help calculate the following probabilities:
      - the probability that both coins are gold
      - the probability that the first coin is gold and second is not
      - the probability that the first coin is silver and the second is gold

- the probability that neither coin is gold

(b) What is the probability that the second coin is gold?

(c) What is the probability that exactly one coin is gold?

9. If $E$ and $F$ are independent events, are $\overline{E}$ and $\overline{F}$ independent? Justify your answer.

10. A bag contains five balls numbered 1, 2, 2, 8 and 8. If three balls are selected at random (without replacement) what is the expected value of the sum of the numbers on the balls.

11. A student is trying to pass calculus. She will stop trying if she pass the course or after three attempts. The probability of passing the course in one attempt is $\frac{1}{3}$. You may assume these attempts are independent. Determine the probability that the student will pass calculus. HINT: If you got 99.9% you are too kind to the student.

12. Consider the following casino game. A person pays \$1 to play. The person tosses a fair coin 4 times.

    - If no heads occur, the player must pay an additional \$2.
    - If one head occurs, the player must pay an additional \$1.
    - If two heads occur the player just looses their initial \$1 paid.
    - If three heads occur the player wins \$3.
    - If four heads occur the players wins \$4.

    What is the expected amount of money the player will win playing this game one time?